Conserved exchange, pair coherence and the physical state boundary

TECT verification-first repository
claim `C6-SPACETIME-SIGNATURE`

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1 Purpose and scope

This checkpoint preserves P1, PAH and EXCH001/002 byte for byte. Its single question is whether an actual preparation–interaction–measurement packet identifies the unchanged exchange law. The overall physical answer is `HOLD_FOR_EVIDENCE`, not an algebra-only replacement PASS. IDEA006/007/008 were retrieved before scoping; the new input is two located primary physical sources. Their mechanism is not full P1 identity.

Keep the side-16 periodic three-torus, $V = 4096$, every separately fixed finite $K \geq 1$, $\Psi \in V_K^3$, $\chi \in V_{2K}$, exact integrals and orthogonal variational projections. Keep the old Hamiltonian and single nonzero sector:

$$H = F_{\mathrm{P1}} + H_\chi + H_{\mathrm{int}}, \quad H_\chi = \int (|\chi|^2/2 + |\chi|^4/4), \quad H_{\mathrm{int}} = -\epsilon \Re Z, \quad \epsilon = 1/16,$$

$$Z = \int \bar{\chi}\, S, \quad S = \sum_{a=1}^{3} \Psi_a^2, \quad Q = N_\psi/2 + N_\chi = \bar{Q} = 11008/27, \quad N_\psi = \int |\Psi|^2, \quad N_\chi = \int |\chi|^2.$$

Time is the original dimensionless τ, with $\{z, \bar{z}\} = -2i$ in orthonormal Fourier coordinates, not an operational clock. No quantum state, Gibbs preparation, dissipation, limiting process, reservoir reset or new two-mode dynamics is installed.

2 Exact diagnostics and the energy ledger

The conversion part of the actual equations is

$$\dot{\Psi}_{\mathrm{int}} = 2i\epsilon P_K(\chi \bar{\Psi}), \qquad \dot{\chi}_{\mathrm{int}} = i\epsilon S.$$

Here $S \in V_{2K}$, and orthogonality removes P_K in the integrated norm pairing. Every pinned P1 term is invariant under the global phase of Ψ: quadratic, density, locking and Class-II currents included. Its contribution to the integrated N_ψ derivative is therefore zero. The reservoir self-energy contributes no N_χ derivative. Consequently

$$\dot{N}_\psi = 4\epsilon \Im Z, \qquad \dot{N}_\chi = -2\epsilon \Im Z, \qquad J := \dot{Q}_\psi = 2\epsilon \Im Z = -\dot{N}_\chi, \qquad \dot{Q} = 0.$$

Both quadratures of Z are weighted-global-$U(1)$ invariant. Use them globally, not $\arg Z$, which fails at $Z = 0$. For $Z \neq 0$, $J = 2\epsilon |Z| \sin(\arg Z)$. The norm fraction $x = Q_\psi/\bar{Q}$ satisfies $\dot{x} = J/\bar{Q}$. It is not yet a measured condensate fraction.

Autonomous Hamiltonian evolution conserves H. Its energy ledger is

$$\dot{F}_{\mathrm{P1}} + \dot{H}_\chi + \dot{H}_{\mathrm{int}} = 0.$$

Population exchange is a norm-charge current, not itself an energy current. In particular $\dot{F}_{\mathrm{P1}} = -\dot{H}_\chi$ requires a separate argument that $\dot{H}_{\mathrm{int}} = 0$; none is assumed. The primary script checks the Poisson factors and three-term bookkeeping symbolically; its generic invariant polynomial test is explicitly not a substitute P1 functional.

3 A phase-blind state does not determine its own drift

Let $q = \bar{Q}/V = 43/432$, $r = q$, and $p = 2(q - r^2) > 0$. The two legal full-field initial points are constant fields

$$\Psi_\pm = \sqrt{p}\, e_1, \qquad \chi_\pm = \pm ir.$$

They have exactly the same Q, N_ψ, N_χ, F_{P1} and H_χ, while $\Re Z_\pm = 0$, so their full energies are equal. But

$$Z_\pm = \mp iVrp, \qquad J_\pm = \mp 2\epsilon Vrp \neq 0.$$

Thus populations, even augmented by $H, Q, \Re Z, |Z|$, do not define a universal autonomous drift reproducing all full preparations. A single population Markov transition law reproducing all projected full Dirac preparations is likewise impossible: the same projected initial point would have two different small-time evolutions. This is an exact initial generator contradiction in the unchanged full regulator. The lock can later rotate e_1; no invariant one-mode subspace is needed or claimed.

This does not exclude a physically justified restricted preparation, memory-dependent dynamics or an error-controlled approximation. Adding $(\Re Z, \Im Z)$ determines the instantaneous population current, but does not prove closure of that larger observable hierarchy.

4 Primary sources and the missing state map

Syassen et al., PRL 99, 033201 (2007), versioned https://arxiv.org/pdf/0704.2155v2, PDF pages 1–2, give a located atom-pair/molecule matrix element (Eq. 1), two-atoms-per-site preparation, fixed-field hold and absorption-imaged entrance-channel population (Fig. 1, Eqs. 2–3). Molecules are invisible to that imaging transition. The source includes field jumps, release/readout and molecular loss; the pair quadrature is not directly imaged. It supplies a physical preparation/readout reference, not the closed EXCH law.

Heinzen et al., PRL 84, 5029 (2000), versioned https://arxiv.org/pdf/cond-mat/0110565v1, PDF pages 1–2, give a pair-conversion operator Hamiltonian (Eq. 2). The passage preceding Eqs. 3–4 explicitly assumes mean replacement and factorization, in a rotating frame; Eq. 7 includes Raman losses. That is a conditional mean-field mechanism, not a proved EXCH state identification.

An elementary reference-only Fock check isolates the issue. For $|A\rangle = |2,0\rangle$, $|M\rangle = |0,1\rangle$ and $|u_\pm\rangle = (|A\rangle \pm i|M\rangle)/\sqrt{2}$,

$$\langle N_a\rangle = 1, \quad \langle N_b\rangle = 1/2, \quad \langle a\rangle = \langle b\rangle = 0, \quad \langle b^\dagger a^2\rangle = \mp i/\sqrt{2}.$$

The last operator preserves $N_a + 2N_b$ although each charged first mean vanishes. Thus the physical pair correlator is not generally the product $\langle b\rangle^*\langle a\rangle^2$. These are exact matrix elements on the physical total-number-two sector, computed with a sufficient finite Fock extension, not an assertion of canonical commutators at truncation edges. This is not quantization of EXCH, a new EXCH preparation or a no-go against every classical representation. It identifies the state-approximation input that may not be hidden behind the shared cubic monomial.

Both source PDFs were hash-checked, read and visually reviewed by the root; their source crosswalk was independently inspected by separate reviewers. Their exact hashes, locators and SATISFIED/FAILED/MISSING rows are in `TECT-EXCH-003-sources-v1.json`. Neither supplies P1's complete three-component fourth-order spatial symbol, internal lock, sextic and Class-II terms with the declared domain, coefficients and units.

The single re-entry input is one source-owned, calibrated *preparation–Hamiltonian–instrument packet*: native carrier/state map; full-law identity or controlled residual on a declared finite time/domain; population and invariant pair-coherence preparation; independent action, length, amplitude, energy and time normalization; actual detector response, covariance and backaction; interaction-energy/control/loss accounting; and an unused intervention testing the same mapping. No observational fit or ad hoc Gibbs state fills these fields here. Literature shows mechanisms, not that such a full EXCH packet is impossible.

5 Devil's-advocate review

1. Wrong sign, factor two or volume: DISMISSED by independent real canonical coordinates with conversion coefficient $\epsilon/\sqrt{V}$ for constant orthonormal Fourier modes.

2. Charge called energy: UPHELD boundary; the interaction-energy derivative remains in the three-term ledger. Units are not calibrated.

3. Constant fields secretly define a replacement two-mode law: DISMISSED; only full-flow initial derivatives enter the contradiction.

4. Phase loss repaired by a selected ensemble: UPHELD prohibition; no physical ensemble or finite closed hierarchy is assumed.

5. Quantum counts identified with classical first means: UPHELD against that import; the fixed-number reference explicitly falsifies unconditional first-mean factorization, not all classical state maps.

6. Measured Rabi exchange certifies full P1 or physical time: UPHELD boundary; source kinetic/carrier/loss/control and state assumptions do not match the full pinned law. No damping term is borrowed.

7. Finite checks certify a limit: UPHELD prohibition. There is no quadrature convergence claim, time sweep, cutoff or volume limit here.

Independent and hostile AI reviewers inspected the analytic boundaries and code; these are not external human referees. External review is invited. Five actually compiled Lean lemmas cover parameterized real algebra and failure of value-only factorization, not full P1 calculus or a physical map. The result is a scoped negative diagnostic and evidence contract, not novel quantum many-body mathematics or an empirical validation of TECT.

6 Result footer

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Result ID: R-580 / C6-SPACETIME-SIGNATURE
Precise statement: Phase-sensitive norm-charge exchange is exact;
  phase-blind universal population closure fails. Physical bridge HOLD.
Scope: Same EXCH finite K>=1, side16 T3, Q=11008/27, time tau.
Dependencies: Frozen EXCH001/002, P1 phase symmetry; two sources
  as mechanism references only, not full physical premises.
Evidence grade: ANALYTIC, EXACT, EXECUTED, CONDITIONAL, INHERITED.
Reproduction command:
  python -X utf8 verification/scripts/exch003_verify.py
Expected output: PASS_SCOPED_AUDIT_PHYSICAL_HOLD.
Falsification gate: Incorrect bracket/projection/full-state witness,
  reference matrix elements or source/state applicability invalidates
  the corresponding conclusion. No claim of exhaustive source absence.
Tier before / after: C6 T1 OPEN / T1 OPEN; T054 gate unchanged.
No-overclaim statement: No physical count/clock, relaxation, closed
  augmented dynamics, PreA, spacetime, QFT, gravity, continuum or TOE.
Next required action: One calibrated full-law/state/instrument packet
  or precise error. No repeat source analogy or automatic successor.
```